

Highlights

MIRAUS: MRI-Informed Residual Adaptation for Single-Frame TRUS Prostate Boundary Measurement

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- Training-only MRI improves single-frame TRUS boundary measurement.
- Utility-guided aggregation uses MRI context without image registration.
- Residual adaptation enables MRI-free automatic and box-assisted use.
- MIRAUS achieved 3.62 mm HD95 across 73 cross-validation patients.
- Model-derived calipers support conventional ellipsoid volume estimation.

MIRAUS: MRI-Informed Residual Adaptation for Single-Frame TRUS Prostate Boundary Measurement

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ARTICLE INFO

Keywords:

Biomedical measurement
health informatics
knowledge distillation
learning using privileged information
prostate boundary measurement
transrectal ultrasound
volume estimation

ABSTRACT

Prostate boundary measurement from transrectal ultrasound (TRUS) provides the in-plane calipers used for conventional ellipsoid volume estimation and downstream prostate-specific antigen density assessment. Speckle noise, acoustic shadowing, weak tissue contrast, probe-induced deformation, and variable localization can degrade these measurements. Although magnetic resonance imaging (MRI) provides superior soft-tissue contrast, it remains impractical as a routine examination input. We formulate MIRAUS under learning using privileged information (LUPI), using weakly aligned MRI only during training and a single TRUS frame at deployment. A utility-supervised module aggregates useful anatomical information from a local 2.5D MRI neighborhood. Residual adaptation then transfers the MRI-conditioned feature increment to a lightweight TRUS student supporting automatic and box-assisted inference. In patient-disjoint five-fold cross-validation on 73 μ RegPro patients (3,885 gland-bearing frames), automatic inference achieved an HD95 of 3.62 mm and a Dice score of 93.20%. In a simulated clinical caliper workflow, the resulting ellipsoid estimates showed a -0.68 -mL bias and a 4.68-mL mean absolute error relative to the reference-caliper ellipsoid. On a separate-center cohort, automatic and standardized box-assisted inference achieved Dice scores of 71.30% and 84.72%, respectively. MIRAUS provides a lightweight route for MRI-informed boundary measurement while preserving automatic and box-assisted single-frame TRUS deployment.

1. Introduction

Prostate cancer represents a major global health burden as the second most frequently diagnosed malignancy in men, accounting for approximately 1.47 million new cases and 397 000 deaths worldwide in 2022 [1]. Given that advanced-stage disease remains largely incurable [2], early detection and long-term surveillance are paramount, with active screening associated with up to a 48.9% reduction in metastatic diagnosis risk [3]. In current urological practice, PSA testing is widely used for initial screening, but its limited specificity contributes to overdiagnosis and unnecessary invasive biopsies [4–6]. Because serum PSA levels are strongly confounded by gland size, prostate volume assessment has become pivotal to calculate PSA density (PSAD)—a key biomarker that differentiates indolent hyperplasia from aggressive malignancy to refine pre-biopsy risk stratification [7, 8].

Transrectal ultrasound (TRUS) is widely used for prostate assessment because it is real-time, portable, and inexpensive. In conventional examination, operators select representative axial and sagittal views and place transverse, anteroposterior, and longitudinal calipers. These three diameters are combined using a prolate-ellipsoid model to

estimate prostate volume [9, 10]. The transverse and anteroposterior measurements depend directly on the perceived gland boundary, so contour errors propagate into the derived volume and PSAD. Speckle noise, acoustic shadowing, weak soft-tissue contrast, probe-induced deformation, view selection, and caliper placement therefore constitute major sources of measurement variation [9, 11].

Magnetic resonance imaging (MRI) provides higher soft-tissue contrast than TRUS and can supply complementary anatomical information for prostate boundary estimation [12]. Multi-observer analyses have also demonstrated close structural correspondence between prostate delineations obtained from MRI and 3D TRUS [13]. MRI is nevertheless unsuitable as a routine input for rapid TRUS measurement because of its cost, acquisition time, and infrastructure requirements [14]. This complementary relationship motivates a learning using privileged information (LUPI) setting: weakly aligned MRI can guide model development, while the deployed measurement system continues to operate from TRUS alone [15].

Automated TRUS segmentation has advanced through convolutional networks, spatial-context modeling, domain adaptation, and promptable medical foundation models [16, 17]. These approaches improve contour estimation, interaction, or computational efficiency, but the deployed prediction remains constrained by the anatomical evidence present in the current TRUS input. Methods that obtain additional spatial information from neighboring frames or complete 3D sweeps also change the acquisition requirement. Existing

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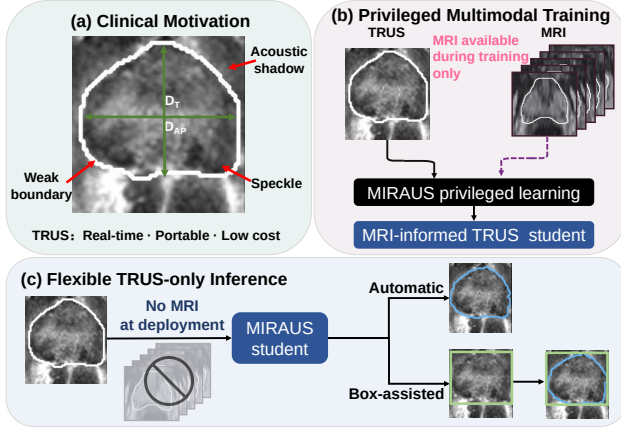


Figure 1: Clinical motivation and deployment concept of MIRAUS. (a) TRUS supports real-time prostate boundary and caliper measurement, but speckle, shadowing, and weak tissue contrast obscure the gland boundary. (b) Weakly aligned MRI provides privileged anatomical information during training. (c) The deployed student performs automatic or box-assisted boundary measurement from a single TRUS frame without MRI.

multimodal and missing-modality approaches provide another route to richer representations, but they commonly assume spatially aligned modalities or transfer cohort-level information. The unresolved problem is how to exploit weakly aligned MRI during training while preserving lightweight, single-frame TRUS measurement at deployment.

To address this problem, we propose MIRAUS, an MRI-informed residual adaptation framework for single-frame TRUS prostate boundary measurement. During privileged teacher training, a coarse relative-depth correspondence defines a local 2.5D MRI neighborhood for each TRUS frame. Utility-supervised aggregation emphasizes the MRI candidates that provide useful anatomical information under this weak alignment. The trained teacher then defines the MRI-conditioned feature increment relative to the original TRUS representation. A lightweight residual adapter learns to estimate this increment from TRUS alone. At deployment, the MRI branch and teacher are removed, and mixed-prompt adaptation allows the same student to operate automatically with a full-image prompt or interactively with a coarse localization box.

The main contributions of this work are summarized as follows¹:

- **Training-time MRI-privileged boundary measurement:** We formulate single-frame TRUS prostate boundary measurement under LUPI, using weakly aligned MRI only during training while retaining lightweight TRUS-only automatic and box-assisted inference.

¹Source code and an interactive case explorer are publicly available at <https://github.com/Zeming-Huang/MIRAUS> and <https://zeming-huang.github.io/MIRAUS/>.

- **Utility-guided anatomical selection and residual transfer:** We introduce utility-supervised aggregation over a local multi-slice MRI neighborhood and distill the resulting MRI-conditioned feature increment into a TRUS student, avoiding dependence on direct pixel-wise alignment.
- **Measurement-oriented validation:** We evaluate boundary distance, in-plane diameter and ellipsoid-volume errors, operator-dependent localization, interaction time, inference latency, anatomical depth, and separate-center acquisition conditions.

2. Related Work

2.1. TRUS Prostate Boundary and Volume Measurement

Conventional TRUS prostate volumetry derives gland volume from transverse, anteroposterior, and longitudinal diameters acquired on selected axial and sagittal views. The diameters are combined using an ellipsoid approximation, providing a rapid measurement that is compatible with routine ultrasound examination [8, 9]. Its reliability depends on the selected imaging plane, the visibility of the gland margin, and the placement of each caliper. Observer studies have consequently reported variation in both TRUS boundary delineation and derived prostate volume [9, 11]. The ellipsoid model also introduces a geometric error when the gland departs from the assumed shape, which can alter the resulting volume and PSAD [7, 8].

Boundary estimation is therefore an integral part of the measurement process. Early automated approaches combined handcrafted appearance descriptors, statistical shape constraints, deformable contours, and level-set optimization to stabilize prostate margins under speckle and acoustic attenuation [18–20]. These methods introduced anatomical regularization into TRUS boundary extraction, but their performance depended on manually designed features and shape assumptions. Their development established the value of boundary geometry for reproducible measurement while motivating data-driven models that can learn more complex TRUS appearances.

2.2. Automated and Promptable TRUS Boundary Estimation

Convolutional networks shifted TRUS boundary estimation from engineered descriptors to learned image representations. Slice-wise models support rapid processing during ultrasound-guided examination, whereas 3D architectures use complete sweeps to model spatial continuity [21–24]. Neighboring-slice input has also been investigated as an intermediate strategy, although additional TRUS slices do not consistently improve segmentation [25]. TRUS-specific architectures further incorporate attention, scanner adaptation, shadow modeling, and domain-aware augmentation [26–30]. These methods improve boundary estimation across acquisition conditions, but spatial-context models commonly

require neighboring frames or complete volumes at inference.

Promptable foundation models provide a complementary interaction mechanism. The Segment Anything Model introduced point-, box-, and mask-conditioned segmentation, and MedSAM adapted this interface to medical images [17, 31]. SAMed and SAM-Med2D support automatic or prompt-assisted medical segmentation, while SAMUS introduces ultrasound-specific local representations [32–34]. Lightweight transformer encoders reduce the computational cost of promptable inference, and volumetric systems such as SegVol and nnInteractive extend interaction to 3D data [35–37]. These developments improve interaction and deployment efficiency, but they still estimate the boundary from the anatomical information available in the deployed TRUS input. Training-time access to a clearer modality offers a separate opportunity to improve single-frame measurement without increasing examination-time imaging requirements.

2.3. Privileged Learning with Weakly Aligned Multimodal Data

Knowledge distillation transfers information from a teacher to a deployable student through output distributions or intermediate representations. Generalized distillation connects this formulation with LUPI, in which auxiliary information is available during training but absent during inference [15, 38]. Modality hallucination similarly trains a single-modality network to reproduce representations learned from a richer training-time input [39].

Medical imaging has applied privileged learning to uncertainty estimation, cross-modal feature supervision, and single-modality deployment [40, 41]. Privileged Modality Distillation transfers complementary information between intravascular imaging modalities, while KD-Net distills a multimodal brain-tumor teacher into a monomodal student [42, 43]. Missing-modality architectures such as RFNet and mmFormer learn multimodal representations that remain usable when an input modality is unavailable, and ProtoKD transfers prototype-level knowledge to a single-modality model [44–46]. These formulations are primarily designed for modalities with dependable spatial correspondence.

MRI guidance for prostate TRUS analysis has also been introduced through deformable registration and MRI-derived anatomical priors [12, 47]. Dense MRI–TRUS correspondence remains difficult because patient positioning, acquisition orientation, slice spacing, and probe pressure alter prostate geometry and image appearance. Direct feature matching under these conditions may associate the current TRUS frame with irrelevant MRI anatomy [48]. Fully unpaired transfer avoids this correspondence requirement, but it discards the coarse within-patient anatomical relationship available in paired acquisitions [49].

Together, these studies show that anatomical priors can stabilize prostate boundaries, multimodal distillation can transfer information available only during training, and promptable models can support different levels of operator

guidance. The remaining challenge is to transfer useful MRI anatomy under coarse MRI–TRUS correspondence while preserving single-frame TRUS deployment. MIRAUS addresses this challenge by aggregating local MRI features through relative-depth correspondence and distilling the resulting MRI-conditioned residual into a TRUS-only student for automatic and box-assisted boundary measurement.

3. Methods

3.1. Measurement Task and Measurement Model

The primary measurement output of MIRAUS is the two-dimensional prostate boundary on a selected TRUS frame. Let $\mathbf{x}_n \in \mathbb{R}^{H \times W}$ denote a single TRUS frame and let $\mathbf{b}$ denote either a full-image prompt or an optional coarse localization box. The deployed student produces a prostate probability map $\hat{\mathbf{p}}_n$, from which the binary mask $\hat{\mathbf{y}}_n$ and its boundary $\hat{C}_n = \partial \hat{\mathbf{y}}_n$ are obtained. The boundary coordinates are initially expressed on the image grid and are converted into physical coordinates using the corresponding pixel spacing.

Two in-plane linear measurements are derived from $\hat{C}_n$. The left–right diameter $\hat{D}_T$ is the transverse extent of the estimated boundary, and the anteroposterior diameter $\hat{D}_{AP}$ is its orthogonal axial extent:

$$\begin{aligned}\hat{D}_T &= \max_{(x,y) \in \hat{C}_n} (x) - \min_{(x,y) \in \hat{C}_n} (x), \\ \hat{D}_{AP} &= \max_{(x,y) \in \hat{C}_n} (y) - \min_{(x,y) \in \hat{C}_n} (y).\end{aligned}\quad (1)$$

where x and y denote physical coordinates in millimetres. The corresponding reference quantities, D_T^{ref} and D_{AP}^{ref} , are derived from the expert reference mask $\mathbf{y}_n$ using the same procedure.

Conventional prostate volume measurement approximates the gland as a prolate ellipsoid [9]:

$$V_{\text{ellip}} = \frac{\pi}{6} D_T D_{AP} D_{SI}, \quad (2)$$

where D_{SI} is the superior–inferior diameter. Diameters are measured in millimetres, and the resulting volume is converted from mm^3 to millilitres.

The retrospective caliper experiment evaluates replacement of the two in-plane measurements on a reference-selected maximal axial cross-section. The model supplies $\hat{D}_T$ and $\hat{D}_{AP}$, while the reference longitudinal diameter D_{SI}^{ref} is held fixed. The resulting assisted estimate is

$$\hat{V}_{\text{ellip}} = \frac{\pi}{6} \hat{D}_T \hat{D}_{AP} D_{SI}^{\text{ref}}. \quad (3)$$

This protocol measures the error introduced when model-derived boundaries replace the two axial calipers within the conventional ellipsoid workflow.

3.2. MIRAUS Framework

We formulate single-frame TRUS prostate boundary measurement under the Learning Using Privileged Information (LUPI) paradigm [15]. Each training sample contains a

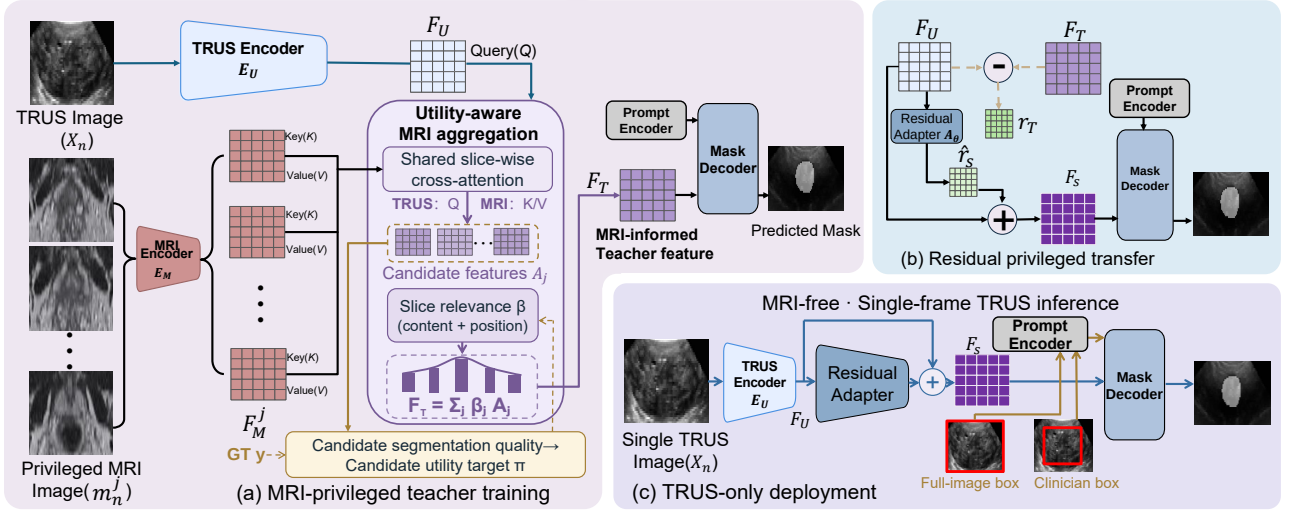


Figure 2: Overview of MIRAUS. (a) During MRI-privileged teacher training, separate frozen encoders extract the TRUS feature F_U and candidate MRI features F_M^j . Each MRI candidate interacts with F_U through shared slice-wise cross-attention. Candidate boundary-estimation quality provides a utility target for the slice weights β , and their weighted aggregation produces the MRI-informed teacher feature F_T . (b) During residual privileged transfer, the frozen teacher defines the MRI-conditioned increment $r_T = F_T - F_U$. The student residual adapter predicts $\hat{F}_S$ from F_U alone and adds it to the TRUS feature to obtain F_S for boundary estimation. (c) At deployment, the MRI encoder and teacher are removed. The same single-frame TRUS student accepts either a full-image box for automatic measurement or a clinician-provided box for assisted measurement.

TRUS frame $\mathbf{x}_n$, its reference mask $\mathbf{y}_n$, and a weakly aligned local MRI neighborhood $\mathcal{M}_n$:

$$\mathcal{D}_{tr} = \{(\mathbf{x}_n, \mathcal{M}_n, \mathbf{y}_n)\}_{n=1}^N. \quad (4)$$

MRI is used by the privileged teacher during training, while the deployed model receives only the current TRUS frame and its prompt.

The overall privileged-learning and deployment pathways are illustrated in Fig. 2. First, the privileged teacher selects and aggregates useful anatomical information from the local MRI neighborhood. Second, a lightweight residual adapter learns to recover the resulting MRI-conditioned feature increment from TRUS alone. Finally, mixed-prompt adaptation prepares the same student for automatic full-image operation and box-assisted measurement. The MRI branch and teacher are removed before deployment.

3.2.1. Modality-Specific Encoding and Local MRI Context

Densely annotated TRUS datasets remain limited, making it difficult to train a high-capacity segmentation model from scratch. Ultrasound examination also favors a compact model that can process individual frames efficiently. We therefore build MIRAUS upon the LiteMedSAM architecture, which contains a TinyViT image encoder, a lightweight prompt encoder, and a mask decoder. The four-stage TinyViT backbone converts each 256×256 input into a 256-channel feature map with a spatial resolution of 64×64 .

TRUS and MRI differ substantially in appearance, so they are processed by separate image encoders. We use two TinyViT encoders with the same architecture but independent modality-specific pretrained weights. Let E_U and E_M

denote the TRUS and MRI encoders, respectively. Given a TRUS frame $\mathbf{x}_n$ and an MRI candidate $\mathbf{m}_n^j$, they produce

$$\mathbf{F}_U = E_U(\mathbf{x}_n), \quad \mathbf{F}_M^j = E_M(\mathbf{m}_n^j). \quad (5)$$

Both image encoders remain frozen during teacher training and student distillation. Training is therefore concentrated on the cross-modal interaction, aggregation, residual adaptation, and mask-prediction modules.

A single MRI slice captures only one cross-section of the prostate. Adjacent slices provide complementary information about gland shape and its gradual variation along the acquisition direction. We therefore extend the privileged MRI context from 2D to 2.5D by constructing a local five-slice neighborhood for each TRUS frame.

We first assign an anchor MRI slice q_n using

$$s_n = \frac{r_n}{R_n - 1}, \quad q_n = \text{round}(s_n(L_n - 1)), \quad (6)$$

where r_n is the TRUS slice index, R_n is the number of TRUS slices, and L_n is the number of MRI slices. The resulting neighborhood is $\mathcal{M}_n = \{(\mathbf{m}_n^j, v_n^j)\}_{j=-2}^2$, where $v_n^j \in \{0, 1\}$ marks whether the candidate lies within the available MRI range. The relative-depth mapping defines a coarse local search region, within which the utility selector identifies the MRI candidates that contribute to the teacher representation.

3.2.2. Utility-Supervised Multi-Slice MRI Aggregation

The local MRI neighborhood provides richer anatomical context, but its candidates are not equally useful for the current TRUS frame. MIRAUS therefore processes each candidate separately and learns a soft aggregation over the

five MRI slices. The aggregation considers TRUS–MRI feature compatibility, relative slice position, and the candidate’s contribution to prostate boundary estimation.

Each MRI candidate first interacts with the TRUS feature through asymmetric multi-head cross-attention. TRUS supplies the query, whereas MRI supplies the keys and values:

$$\mathbf{A}_j = \text{LN}_2 \left[\mathbf{F}_U + W_O \text{MHA} \left(W_Q \text{LN}_1(\mathbf{F}_U), W_K \text{LN}_1(\mathbf{F}_M^j), W_V \text{LN}_1(\mathbf{F}_M^j) \right) \right]. \quad (7)$$

This interaction keeps the current TRUS frame as the representation anchor while allowing each MRI candidate to provide complementary anatomical information.

The five interacted candidates must then be weighted. Learnable mappings first generate normalized feature descriptors for the TRUS frame and each MRI slice:

$$\mathbf{d}_U = \frac{g_U [\rho(\mathbf{F}_U, \mathbf{b}_\Omega)]}{\|g_U [\rho(\mathbf{F}_U, \mathbf{b}_\Omega)]\|_2}, \quad \mathbf{d}_M^j = \frac{g_M [\rho(\mathbf{F}_M^j, \mathbf{b}_\Omega)]}{\|g_M [\rho(\mathbf{F}_M^j, \mathbf{b}_\Omega)]\|_2}. \quad (8)$$

Here, $\rho(\cdot)$ concatenates global, prompt-region, and prompt-boundary pooled features, while g_U and g_M are modality-specific projections. During teacher training, $\mathbf{b}_\Omega$ spans the full image.

The content compatibility between the current TRUS frame and MRI candidate j is measured by cosine similarity. The selector combines this similarity with a learned relative-position bias and a weak Gaussian prior:

$$a_j = \alpha \langle \mathbf{d}_U, \mathbf{d}_M^j \rangle + u_j + \lambda_p \log p_j, \quad (9)$$

where $\alpha > 0$ controls the contribution of content similarity, u_j is a learned relative-position bias, and $p_j \propto \exp(-j^2/2\sigma_p^2)$. The spatial prior favors nearby candidates without forcing the assigned center to receive the largest weight.

Let $\mathcal{V}_n = \{j \mid v_n^j = 1\}$ denote the valid MRI candidates. Their normalized aggregation weights are

$$\beta_j = \begin{cases} \frac{\exp(a_j)}{\sum_{k \in \mathcal{V}_n} \exp(a_k)}, & j \in \mathcal{V}_n, \\ 0, & j \notin \mathcal{V}_n. \end{cases} \quad (10)$$

Feature compatibility alone cannot determine whether an MRI candidate improves the boundary estimate for the current TRUS frame. We therefore supervise the selector using candidate-specific boundary utility. A lightweight auxiliary head C_{cand} predicts a mask from each interacted feature $\mathbf{A}_j$, and the corresponding segmentation error is

$$e_j = \mathcal{L}_{\text{Dice}}(C_{\text{cand}}(\mathbf{A}_j), \mathbf{y}_n) + \mathcal{L}_{\text{BCE}}(C_{\text{cand}}(\mathbf{A}_j), \mathbf{y}_n). \quad (11)$$

A lower error indicates that the corresponding MRI candidate provides more useful information for estimating the

current prostate boundary. The errors are converted into the utility distribution

$$\pi_j = \begin{cases} \frac{\exp(-e_j/T_u)}{\sum_{k \in \mathcal{V}_n} \exp(-e_k/T_u)}, & j \in \mathcal{V}_n, \\ 0, & j \notin \mathcal{V}_n, \end{cases} \quad (12)$$

where T_u controls the sharpness of the target distribution. The utility target is detached before it supervises the predicted aggregation weights:

$$\mathcal{L}_{\text{util}} = D_{\text{KL}}(\text{sg}(\boldsymbol{\pi}) \parallel \boldsymbol{\beta}), \quad (13)$$

where $\text{sg}(\cdot)$ denotes the stop-gradient operator. This supervision teaches the selector to prioritize candidates according to their boundary-estimation utility, complementing the visual similarity and relative-position terms.

Finally, the valid candidate features are combined using the learned weights:

$$\mathbf{F}_T = \sum_{j \in \mathcal{V}_n} \beta_j \mathbf{A}_j. \quad (14)$$

The resulting teacher feature preserves the TRUS representation as its base while incorporating the MRI information selected for the current frame.

3.2.3. Residual Privileged Adaptation

After teacher training, the complete teacher is frozen. It generates target features during student training, with no gradients propagated into any teacher component.

We express the teacher representation relative to the shared frozen TRUS feature:

$$\mathbf{r}_T = \text{sg}(\mathbf{F}_T - \mathbf{F}_U). \quad (15)$$

The student uses a lightweight residual adapter A_θ to estimate this increment from $\mathbf{F}_U$ alone:

$$\hat{\mathbf{r}}_S = A_\theta(\mathbf{F}_U), \quad \mathbf{F}_S = \mathbf{F}_U + \hat{\mathbf{r}}_S. \quad (16)$$

The adapter contains two parallel convolutional branches whose outputs are summed. The final layer of one branch is zero-initialized to stabilize the beginning of student training.

Because both representations share the same frozen TRUS feature, SmoothL1 matching of the complete features is equivalent to residual matching under identical spatial averaging. This formulation makes the student-predicted correction explicit.

The predicted correction is aligned with the teacher-defined increment using

$$\mathcal{L}_{\text{priv}} = \text{SmoothL1}(\hat{\mathbf{r}}_S, \mathbf{r}_T). \quad (17)$$

The loss is averaged over all channels and spatial positions of the feature map.

Together with segmentation supervision, the complete student objective is

$$\mathcal{L}_S = \mathcal{L}_{\text{seg}} + 2\mathcal{L}_{\text{priv}}, \quad (18)$$

where $\mathcal{L}_{\text{seg}}$ combines Dice, binary cross-entropy, and mask-quality regression losses. Once student training is complete, neither $\mathbf{F}_T$ nor $\mathbf{r}_T$ is required for inference.

3.3. Training Objectives and Mixed-Prompt Adaptation

Training comprises three sequential stages. First, the privileged teacher receives the TRUS frame, its local MRI neighborhood, and the reference mask. Both image encoders remain frozen, while the cross-modal interaction, MRI aggregation, candidate head, and teacher-side segmentation components are optimized. The candidate head receives direct segmentation supervision, and its detached candidate errors define the utility target π that supervises the aggregation weights β through $\mathcal{L}_{\text{util}}$.

Second, the trained teacher is frozen and used only for forward computation. It supplies the residual target $\mathbf{r}_T$, while the student residual adapter predicts $\hat{\mathbf{r}}_S$ from $\mathbf{F}_U$. The student representation $\mathbf{F}_S$ is optimized using the inherited segmentation objective together with $2\mathcal{L}_{\text{priv}}$. No gradients are propagated into the teacher or MRI encoder during this stage.

Third, mixed-prompt adaptation prepares the same student for automatic and box-assisted measurement. During transrectal acquisition, the probe is positioned close to the gland, and the prostate commonly occupies a large proportion of the image. A full-image box therefore provides a loose localization prompt for automatic operation:

$$\mathbf{b}_\Omega = [0, 0, W - 1, H - 1]. \quad (19)$$

Image quality, boundary ambiguity, and the preferred level of operator control vary across examinations. Mixed-prompt adaptation therefore combines full-image rehearsal with perturbed localization boxes $\mathbf{b}_n^{\text{loc}}$. This strategy retains the full-image behavior while teaching the prompt encoder and mask decoder to use coarse localized guidance. During this stage, the TRUS encoder and residual adapter remain frozen, while the prompt encoder and mask decoder are optimized using $\mathcal{L}_{\text{seg}}$.

At deployment, the student prediction is

$$\hat{\mathbf{p}}_n = \mathcal{S}_\theta(\mathbf{x}_n, \mathbf{b}), \quad \mathbf{b} \in \{\mathbf{b}_\Omega, \mathbf{b}_n^{\text{loc}}\}. \quad (20)$$

Both operating modes use the same single-frame TRUS student. MRI and the privileged teacher are absent from the deployed measurement path.

3.4. Experimental protocol

3.4.1. Datasets and Evaluation Cohorts

We evaluated MIRAUS across two distinct clinical cohorts to rigorously assess both volumetric accuracy and zero-shot domain generalization. The internal cohort comprises the public μRegPro dataset [50], containing paired, unaligned MRI and TRUS acquisitions from 73 patients. Model performance was evaluated using patient-disjoint 5-fold cross-validation. Pooling out-of-fold predictions across all 73 patients yielded 3,885 gland-bearing TRUS frames for 3D patient-macro analysis. To evaluate direct cross-center transferability under real-world domain shifts, we further established an independent external test cohort consisting

Table 1

Summary of Datasets and Evaluation Protocols

Dataset	Protocol	Frames (Pts)	Modalities
μRegPro	5-Fold Patient CV	3,885 (73)	TRUS + MRI (<i>Train</i>)
China–Japan Friendship Hospital dataset	External Test	327 (24)	TRUS Only

of 327 gland-bearing 2D TRUS frames collected at China–Japan Friendship Hospital. The collection and use of this external cohort were approved by the Science and Technology Ethics Committee of the University of Chinese Academy of Sciences (No. UCASSTEC26-003), with written informed consent obtained from all participants. This external dataset lacks paired MRI acquisitions and was used strictly to evaluate the zero-shot deployment of the trained TRUS-only student model. Dataset statistics and evaluation settings are summarized in Table 1.

All models were implemented in PyTorch and trained on NVIDIA GPUs. For MIRAUS, TRUS and MRI inputs were resized to 256×256 . Both TRUS and MRI TinyViT backbones remained frozen throughout optimization to leverage foundation vision priors. Training proceeded in three sequential stages using the AdamW optimizer with an initial learning rate of 5.5×10^{-4} , weight decay of 8.1×10^{-5} , and cosine learning rate scheduling. In Stage 1, the privileged teacher was trained for 20 epochs (512 sampled frames per epoch, batch size 2), aggregating multi-slice MRI candidates via utility-supervised KL divergence. In Stage 2, the teacher was frozen, and the student was distilled for 20 epochs using the objective in Eq. (18). In Stage 3, the student underwent 5 epochs of mixed-prompt adaptation, during which prompts were randomly sampled from standard full-image prompts $\mathbf{b}_\Omega$ and noise-perturbed bounding boxes $\mathbf{b}_n^{\text{loc}}$.

3.4.2. Evaluation Metrics and Retrospective Volumetric Protocol

For the μRegPro cohort, metrics were calculated from predictions pooled within each patient and summarized across all 73 patients with equal weighting. HD95 was measured in millimetres on the 0.8-mm isotropic grid and in pixels for the external cohort, which lacked validated pixel-spacing metadata. Standardized box-assisted comparisons used identical reference-mask-derived boxes across methods; we further conducted experiments with bounding boxes drawn by multiple clinical observers in Section 5.2. Segmentation performance was assessed via Dice, IoU and HD95, while volume error was quantified by MAE in millilitres. The retrospective caliper experiment followed Section 3.1, comparing model-assisted ellipsoid estimates with both reference-caliper ellipsoid volumes and voxel-count reference volumes. Reference volumes were computed from the ground-truth masks using the isotropic voxel spacing; the reference superior-inferior diameter was their full longitudinal extent.

4. Results

4.1. Main Benchmark Comparison and Cross-Site Generalization

On the public μ RegPro dataset, MIRAUS achieved the best performance among the evaluated baselines across 73 patients (3,885 gland-bearing TRUS frames). By leveraging privileged multi-slice MRI guidance during training, the network operates in a lightweight, TRUS-only manner at inference, avoiding MRI processing and cross-modal alignment at deployment.

As summarized in Table 2, MIRAUS achieves superior segmentation performance across the evaluated settings. Standard fully automatic single-modality architectures (Rows 1–4) exhibit limited spatial accuracy due to standard TRUS contrast constraints, with the top performer (nnU-Net) achieving 84.45% Dice. Crucially, despite being trained on paired MRI-TRUS data and specifically engineered to accommodate missing modalities during inference, architectures such as RFNet and mmFormer (Rows 5–6) still yield suboptimal TRUS-only results (Dice of 76.46% and 69.94%). This indicates that simply training multi-modal architectures to tolerate missing modalities remains insufficient for prostate segmentation, where severe imaging artifacts demand explicit privileged feature learning rather than passive multi-modal robustness.

By framing cross-modal learning through privileged information distillation (LUPI), MIRAUS (automatic, Row 8) avoids any multi-modal inference dependency altogether. It significantly outperforms existing feature-prototypical distillation strategies (ProtoKD in Row 7, 79.99% Dice), advancing automatic performance to 93.20% Dice, 87.41% IoU, and reducing HD95 boundary error to 3.62 mm. Furthermore, when evaluated against interactive, box-prompted foundation models (Rows 9–13), MIRAUS (box-assisted, Row 14) achieves competitive internal accuracy (92.56% Dice) and the highest external box-assisted Dice among the evaluated methods (84.72%). Notably, even in its automatic full-image-prompt mode, MIRAUS (93.20% Dice) surpasses all box-assisted foundation models on the internal cohort (e.g., SAMUS box-only at 92.07% and LiteMedSAM at 88.82%), confirming that internalizing privileged MRI structural priors provides richer contextual awareness than manual bounding box constraints alone.

To visually inspect volumetric shape and slice-level boundary precision, Fig. 3 compares MIRAUS with representative automatic (nnU-Net) and prompt-guided (MedSAM) baselines on the internal cohort across the apex, mid-gland, and base. In the illustrated cases, nnU-Net shows shape irregularity and boundary over-estimation where prostate boundaries blend into adjacent structures. MedSAM remains localized by its box prompt but shows mid-gland under-segmentation and basal boundary leakage. In contrast, the MIRAUS contours more closely follow the reference annotations across the displayed anatomical zones. Extended visual comparisons across 15 anonymized cases are available in

our online result viewer (<https://zeming-huang.github.io/MIRAUS/>).

On the separate-center external cohort (China–Japan Friendship Hospital dataset), fully automatic MIRAUS achieved 71.30% Dice, outperforming the evaluated non-prompted automatic baselines. Under the standardized reference-mask-derived box condition shared by all box-prompted methods, MIRAUS achieved 84.72% Dice, 73.91% IoU, and an HD95 boundary error of 32.05 pixels. This 13.42-point Dice boost underscores the value of human-in-the-loop interaction for cross-center deployment. Minimal user prompting provides critical spatial anchoring to mitigate localized imaging artifacts and scale variations, while introducing only modest manual burden.

As illustrated by the qualitative slice comparisons in Fig. 4, standard automatic baselines like nnU-Net (purple contours) collapse under cross-institution domain shift, resulting in severe under-segmentation (Dice as low as 0.283 in Case A and 0.617 in Case B) or boundary leakage into neighboring tissues (Case C). Interactive foundation models like LiteMedSAM (blue contours) maintain spatial localization via user bounding boxes but produce overly smooth, schematic boundaries that fail to capture intricate patient-specific prostate contours. In contrast, MIRAUS (red contours) exhibits improved anatomical alignment across varying patient geometries, achieving comparatively well-delineated boundaries with per-slice Dice scores of 0.912 in Case A and 0.899 in Case C. These observations indicate that leveraging privileged multi-slice MRI structural priors can help alleviate localized boundary shifts when generalizing to external clinical centers.

4.2. Ablation Study

We systematically validate the core methodological components of MIRAUS: utility-supervised multi-slice MRI aggregation in the teacher, and the residual privileged distillation scheme in the student. Table 3 summarizes the complete ablation hierarchy evaluated in TRUS-only deployment mode alongside the fine-tuned LiteMedSAM baseline.

Efficacy of Utility-Supervised MRI Aggregation Table 3 evaluates teacher aggregation strategies against the fine-tuned LiteMedSAM baseline (88.82% Dice, 6.62 mL Volume MAE).

Even unguided aggregation strategies (A1 and A2) achieve substantial performance gains over the foundation baseline (improving Dice to 92.07% and 91.36%, and reducing Volume MAE to 4.41 mL and 4.84 mL, respectively). This baseline gain confirms that aggregating multi-slice privileged MRI context in the teacher is fundamentally beneficial for cross-modal transfer. However, unguided feature fusion remains sub-optimal because it does not explicitly condition the teacher’s representations on downstream student transferability. In contrast, introducing explicit segmentation utility supervision (A3) encourages the teacher to synthesize feature representations specifically optimized for student residual learning. This targeted guidance yields further comprehensive improvements across all

Table 2

Segmentation performance under five-fold cross-validation and external cohort

#	Method	Params (M)	μ RegPro			External cohort		
			Dice (%) $\uparrow$	IoU (%) $\uparrow$	HD95 (mm) $\downarrow$	Dice (%) $\uparrow$	IoU (%) $\uparrow$	HD95 (pixel) $\downarrow$
Fully automatic methods (TRUS-only)								
1	3D U-Net [23]	4.81	77.39	66.78	10.67	—	—	—
2	UNETR++ [51]	42.95	77.52	67.21	11.56	—	—	—
3	nnU-Net [24]	24.27	84.45	75.95	8.85	52.72	39.95	83.76
4	SAMed [32]	91.35	75.07	65.76	37.10	66.53	54.22	39.18
5	RfNet [44]	8.69	76.46	64.68	36.63	—	—	—
6	mmFormer [45]	36.65	69.94	56.67	72.74	—	—	—
Privileged distillation methods (MRI-guided training, TRUS-only inference)								
7	ProtoKD [46]	3.27	79.99	70.04	23.14	—	—	—
8	MIRAUS (automatic)	15.92	93.20	87.41	3.62	71.30	57.25	59.20
Box-assisted methods (Interactive / Box-prompted inference)								
9	LiteMedSAM [17]	9.79	88.82	80.29	3.79	83.96	72.57	35.67
10	SAM-Med2D [33]	271.24	85.78	76.82	13.66	44.43	32.14	99.92
11	SegVol (no zoom-in) [36]	180.89	82.79	73.81	17.89	—	—	—
12	nnInteractive [37]	383.50	78.32	65.37	25.73	76.99	63.97	62.30
13	SAMUS box-only [34]	133.32	92.07	85.71	8.33	—	—	—
14	MIRAUS (box-assisted)	15.92	92.56	86.36	4.19	84.72	73.91	32.05

Note: Missing entries (—) indicate methods not evaluated on the external cohort under the corresponding experimental setting; such methods rely on volumetric or multimodal inputs that are unavailable in the external 2D TRUS cohort. External HD95 values are reported in pixels because the exported images lack validated physical pixel-spacing metadata.

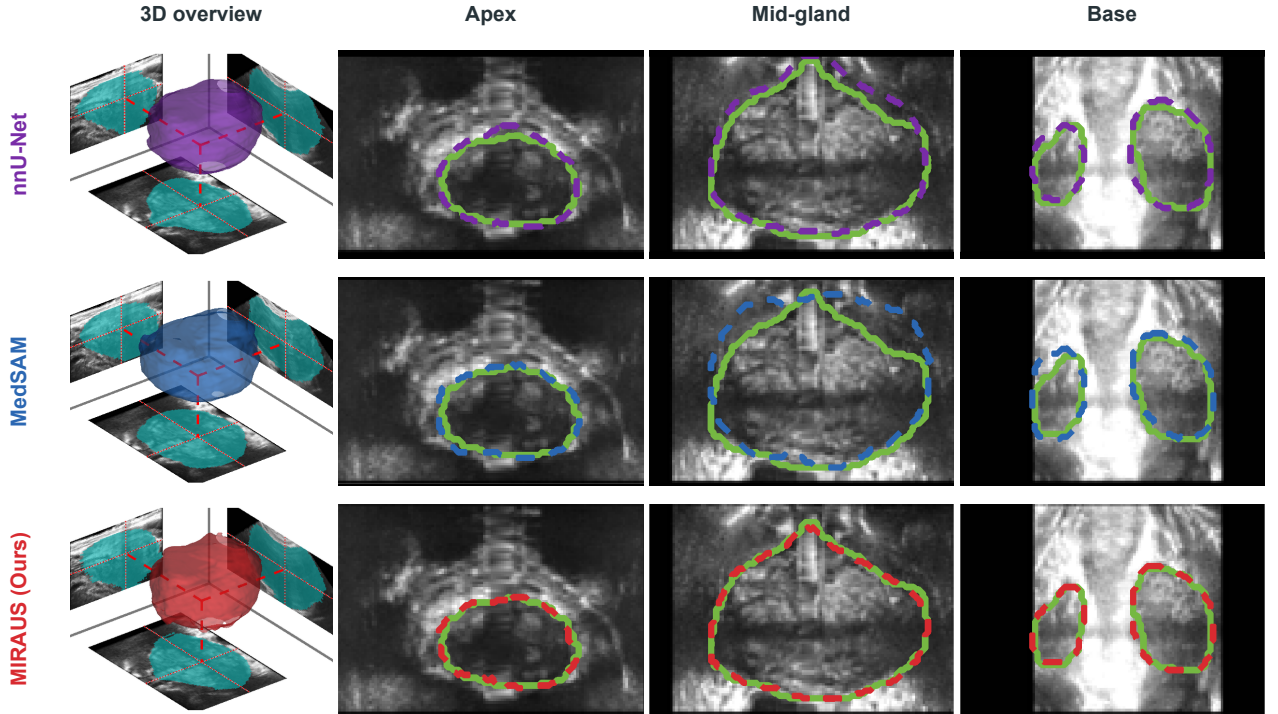


Figure 3: Qualitative 3D volumetric and 2D multi-slice segmentation comparison on the public μ RegPro dataset across Apex, Mid-gland, and Base regions. From top to bottom: nnU-Net (purple), MedSAM (blue), and the proposed MIRAUS (red), with ground-truth contours delineated in solid green.

evaluation metrics, pushing student performance to 93.20% Dice, reducing HD95 to 3.62 mm, and advancing volume estimation precision to 3.61 mL Volume MAE—achieving a reduction of over 45% in volumetric error over fine-tuned LiteMedSAM.

Efficacy of Residual Privileged Distillation Regarding the student distillation objective using the proposed utility-guided teacher (A3), conventional logit distillation B2 proves effective over the non-distilled TRUS baseline B1

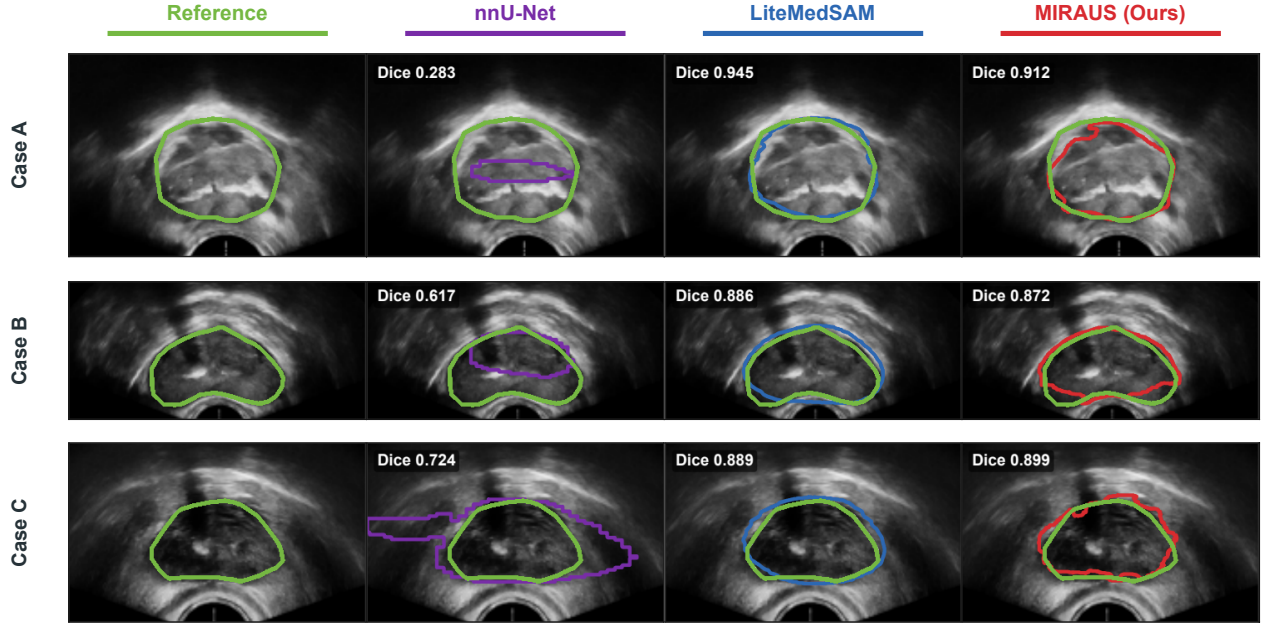


Figure 4: Qualitative comparison of segmentation performance on the external verification cohort (China–Japan Friendship Hospital dataset).

Table 3
Ablation study in TRUS-only deployment mode.

ID	Teacher Agg.	Student KD	Dice (%) $\uparrow$	IoU (%) $\uparrow$	HD95 (mm) $\downarrow$	Vol MAE (mL) $\downarrow$
–	–	LiteMedSAM (Fine-tuned)	88.82 ± 5.41	80.29 ± 7.92	3.79 ± 2.15	6.62 ± 6.10
A1	Uniform Avg.	Residual ($\mathcal{L}_{\text{priv}}$)	92.07 ± 3.89	85.53 ± 6.35	4.44 ± 2.07	4.41 ± 4.39
A2	Unguided Soft	Residual ($\mathcal{L}_{\text{priv}}$)	91.36 ± 4.83	84.43 ± 7.66	4.55 ± 2.42	4.84 ± 5.34
A3	Util.-Supervised	Residual ($\mathcal{L}_{\text{priv}}$)	93.20 ± 3.00	87.41 ± 5.05	3.62 ± 1.81	3.61 ± 4.00
B1	Util.-Supervised	None (TRUS-only)	91.94 ± 5.20	85.45 ± 7.84	4.26 ± 2.23	4.22 ± 5.47
B2	Util.-Supervised	Logit KD	92.86 ± 3.37	86.84 ± 5.60	3.70 ± 1.86	3.64 ± 3.79
B3	Util.-Supervised	Residual ($\mathcal{L}_{\text{priv}}$)	93.20 ± 3.00	87.41 ± 5.05	3.62 ± 1.81	3.61 ± 4.00

(91.94% Dice, 4.22 mL Volume MAE), advancing performance to 92.86% Dice and 3.64 mL Volume MAE. This demonstrates that transferring overall output probability distributions helps regularize the lightweight student.

Crucially, our proposed residual privileged distillation objective B3 ($\mathcal{L}_{\text{priv}}$, weighted by $\lambda = 2$) outperforms conventional KD across all metrics, reaching the overall top performance of 93.20% Dice, 87.41% IoU, 3.62 mm HD95, and 3.61 mL Volume MAE. While conventional KD only transfers coarse output logit distributions, B3 directly supervises the student to predict the fine-grained incremental anatomical representation $\mathbf{r}_T$ provided by privileged MRI. This formulation enables the lightweight TRUS student to effectively internalize cross-modal structural priors and achieve maximal performance gain (+4.38% Dice and +7.12% IoU over fine-tuned LiteMedSAM).

4.3. Analysis of Privileged MRI Context and Modality Dependence

To understand how privileged multi-slice MRI representations empower cross-modal distillation, we evaluated the effect of training-time MRI context size and verified the teacher’s functional dependence on MRI input.

Impact of Privileged MRI Neighborhood Size During privileged training, the teacher aggregates a spatial neighborhood of multi-slice MRI representations to construct rich anatomical context for the TRUS-only student. As detailed in Table 4(a), expanding the training-time MRI context from a single slice to 3 and 5 slices progressively enhances student inference performance, with the 5-slice configuration achieving the highest accuracy (93.20% Dice and 87.41% IoU). However, further expanding the context window to 7 slices leads to a marked performance degradation (87.60% Dice, representing a 5.60 pp drop from the reference). This

Table 4
Analysis of Privileged Teacher MRI Utilization

(a) Effect of Privileged MRI Context Size			
MRI Slices	Dice (%) $\uparrow$	Δ Dice (pp)	IoU (%) $\uparrow$
1 Slice	90.84	-2.36	83.33
3 Slices	92.15	-1.05	85.55
5 Slices	93.20	Reference	87.41
7 Slices	87.60	-5.60	78.39

(b) Counterfactual Teacher Dependence on Privileged MRI Input			
MRI Condition	Dice (%) $\uparrow$	Δ Dice (pp)	SD
Matched MRI	86.40	Reference	± 3.94
Cross-Patient Mismatch	83.70	-2.69	± 4.10
Zeroed MRI Input	64.81	-21.59	± 12.70

indicates that an excessively wide spatial window introduces weakly correlated out-of-plane features that dilute essential local anatomical guidance. Thus, a compact 5-slice MRI neighborhood provides the best-performing context size among the evaluated configurations.

Counterfactual Verification of Teacher MRI Dependence To examine the teacher network’s dependency on privileged MRI cues rather than functioning as a redundant feature extractor, we conducted counterfactual input interventions across TRUS frames (Table 4(b)). Under standard matched privileged MRI input, the frozen teacher achieves a baseline segmentation performance of 86.40% Dice.

When evaluating input sensitivity, completely zeroing out the MRI feature representation serves as an out-of-distribution intervention on the frozen teacher, causing a severe performance degradation of 21.59% (dropping to 64.81% Dice). This substantial decrease indicates that the privileged teacher relies heavily on cross-modal representations to construct its residual supervision targets. Furthermore, replacing the patient-matched MRI with cross-patient scans results in a moderate performance decrease of 2.69% (83.70% Dice). This behavior suggests that the privileged branch utilizes two complementary components: the drop under cross-patient mismatch reflects the omission of subject-specific anatomical boundary cues, whereas the residual 83.70% Dice score demonstrates that the architecture still extracts transferable, general prostate structural priors from unmatched MRI slices.

5. Discussion

5.1. Mechanistic Insights into Privileged Feature Distillation

In this study, we presented MIRAUS, a lightweight, prompt-adaptable segmentation framework that leverages privileged multi-slice MRI representations during training to elevate TRUS-only boundary estimation. By reframing cross-modal guidance within a Learning Using Privileged Information (LUPI) paradigm, MIRAUS simultaneously tackles two critical challenges in intraoperative imaging:

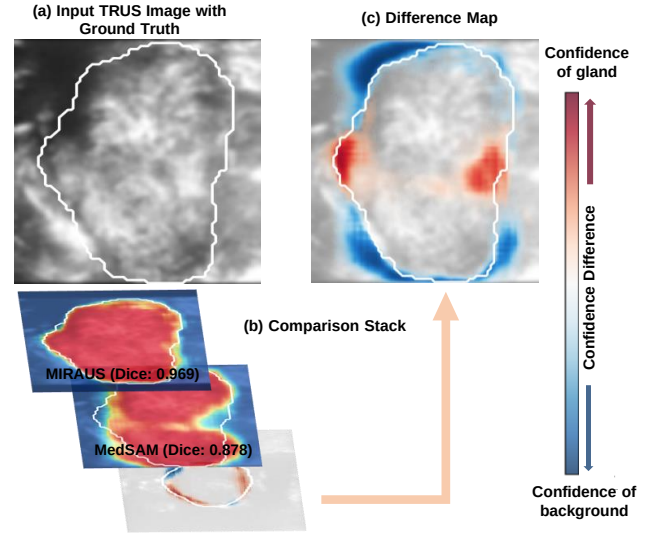


Figure 5: Slice-level confidence difference analysis between MIRAUS and MedSAM. (a) Input TRUS frame overlaid with ground truth (white contour). (b) Predicted probability maps for MIRAUS and MedSAM. (c) Difference map showing region-wise confidence divergence (red: higher MIRAUS confidence for prostate tissue; blue: higher confidence for background).

overcoming the intrinsic anatomical ambiguity of TRUS through privileged MRI representations, and ensuring high clinical usability via ultra-fast, prompt-adaptable TRUS-only deployment.

The key design choice of MIRAUS stems from addressing real-world clinical bottlenecks: preoperative MRI availability is inconsistent, and pixel-level 3D cross-modal registration is often unfeasible due to soft-tissue deformation. To circumvent these constraints, MIRAUS relies on a flexible depth-aware mapping coupled with utility-supervised aggregation, effectively leveraging privileged MRI guidance during training without requiring dense deformable cross-modal registration.

Our diagnostic analyses reveal a fundamental distinction between standalone teacher performance and student transferability. Unsupervised soft aggregation allows the privileged teacher to maximize its own segmentation accuracy on paired MRI-TRUS inputs; however, it fails to optimize the representations for cross-modal distillation. Introducing explicit segmentation utility supervision regularizes the teacher aggregation weights, forcing the teacher to construct anatomical representations specifically tailored for residual estimation by the student. Consequently, the lightweight TRUS student effectively learns the MRI-conditioned feature increment $\mathbf{r}_T$, closing the performance gap against compute-heavy foundation models without incurring any inference-time multi-modal overhead.

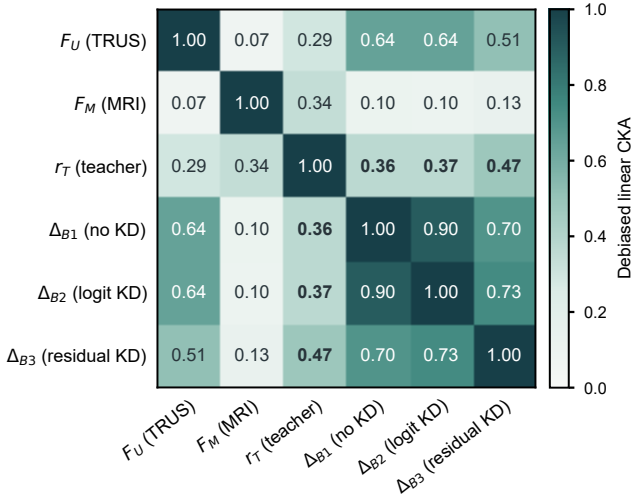


Figure 6: Centered Kernel Alignment (CKA) similarity matrix across internal representations. Mean CKA scores evaluate feature-space alignment between single-modality sources (F_U, F_M), the teacher-defined residual target (r_T), and student-learned task corrections without distillation (Δ_{B1}), with logit distillation (Δ_{B2}), and with residual distillation (Δ_{B3}).

To directly visualize how privileged MRI distillation refines local boundary decision-making compared to baseline foundation models, Fig. 5 provides a slice-level confidence difference analysis between MIRAUS and MedSAM on a representative case. As depicted in Fig. 5(b), MIRAUS achieves a significantly higher local Dice score (0.969 vs. 0.878) with a sharply defined probability boundary adhering to the reference contour. The resulting difference map (Fig. 5(c)) reveals two key mechanisms behind this improvement: the prominent red regions along the lateral prostate boundaries indicate that MIRAUS assigns substantially higher gland confidence, successfully recovering tissue regions overlooked by MedSAM’s under-segmentation; simultaneously, the blue regions near the peripheral borders reflect increased background confidence, effectively eliminating the spurious boundary leakage and halo artifacts common in interactive foundation models. This pixel-level divergence provides evidence that distilling MRI-conditioned residual features (r_T) equips the TRUS student with sharper spatial discrimination, directly accounting for the marked reduction in Hausdorff distance (HD95).

To determine whether residual distillation produced the intended feature-space adaptation beyond output-level logit matching, we analyzed the internal representations using debiased linear Centered Kernel Alignment (CKA) [52] (Fig. 6). The analysis included the TRUS base feature F_U , the MRI feature F_M , the teacher-defined increment r_T , and the student corrections learned without distillation (Δ_{B1}), with logit distillation (Δ_{B2}), and with residual distillation (Δ_{B3}). CKA was computed independently within each cross-validation fold and then averaged across the five folds.

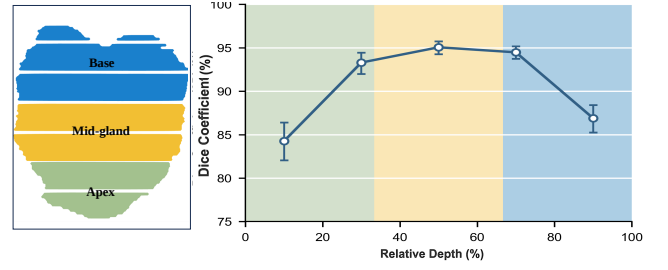


Figure 7: Segmentation performance across relative axial prostate depths. (Left) Anatomical region categorization across five equal depth intervals. (Right) Patient-macro mean regional 3D Dice scores (%) across relative axial depths. Error bars indicate 95% confidence intervals obtained from 20,000 patient-level bootstrap resamples.

The mean CKA matrix confirms that F_M retains a distinct representation geometry from F_U (0.07) and the student corrections (0.10–0.13). Crucially, while the baseline (Δ_{B1}) and logit-distilled (Δ_{B2}) corrections yield similar alignment with the teacher target r_T (CKA = 0.36 and 0.37, respectively), our residual distillation (Δ_{B3}) elevates this feature alignment to 0.47, achieving a consistent improvement across all five cross-validation folds (mean gain of +0.108). Simultaneously, the Δ_{B3} correction reduces reliance on the raw TRUS feature F_U (CKA drops from 0.64 to 0.51). This coordinated shift confirms that residual distillation successfully redirects the learned correction away from the raw TRUS baseline and toward the privileged cross-modal increment, providing internal feature-space evidence for the qualitative boundary refinements observed in practice.

5.2. Clinical Feasibility, Efficiency, and Deployment Analysis

To assess the practical clinical viability of MIRAUS for TRUS measurement workflows, we evaluated four operational dimensions: computational efficiency, anatomical depth sensitivity, operator localization robustness under imperfect bounding boxes, and agreement with standard clinical volume measurements.

Model Complexity and Runtime Latency As summarized in Table 2, MIRAUS balances segmentation performance and computational footprint. With 15.92 M parameters, MIRAUS is lighter than foundation models such as SAM-Med2D (271.24 M), SegVol (180.89 M), and nnInterActive (383.50 M). The TRUS-only student requires no privileged MRI processing at deployment. Inference on a single 2D TRUS frame takes under 15 ms on an NVIDIA GeForce RTX 5070 Ti GPU. In box-assisted mode, manual box placement required 4.12–5.60 s per frame across observers, and the complete measured workflow, including inference and post-processing, required 4.36–5.85 s per frame. These retrospective timing results verify the model’s low-latency processing capability and lay a solid foundation for further prospective workflow validation.

Anatomical Zonal Sensitivity along Axial Depth To examine depth-dependent segmentation performance across anatomical sub-regions, each patient's gland-bearing slice sequence was normalized to a relative axial depth of 0%–100% and categorized into five equal intervals (Fig. 7). Within each interval, regional 3D Dice was calculated per patient by pooling foreground counts across constituent slices and averaged across patients with equal weighting. Uncertainty in the estimated mean performance was quantified using 95% confidence intervals derived from 20,000 patient-level bootstrap resamples. As shown in Fig. 7, MIRAUS achieves its peak accuracy in the central depth interval (40%–60%, mean regional Dice 95.06%; 95% CI: 94.27%–95.75%). Performance naturally declines toward the terminal apical (0%–20%, 84.27%) and basal (80%–100%, 86.89%) intervals. This pattern aligns with clinical expectations, as mid-gland slices offer well-defined anatomical contours, whereas apical and basal regions suffer from severe acoustic shadowing, low signal-to-noise ratios, and ill-defined prostate-bladder/rectal interfaces. Nevertheless, as qualitative comparisons confirm (Fig. 3), privileged MRI distillation significantly mitigates boundary degradation in these challenging terminal slices, avoiding the severe boundary collapse common in non-privileged baselines.

Operator-Assisted Localization and Robustness to Box Imperfections On the separate-center TRUS cohort, fully automatic inference showed performance degradation under variations in field of view, imaging gain, and scale. We therefore evaluated box-assisted inference as a complementary operating mode. To quantify robustness to variable operator-supplied bounding boxes, ground-truth segmentations annotated by a senior urological imaging expert were adopted as the reference standard. Two clinical observers with differing levels of experience independently placed bounding boxes on a representative evaluation subset of external-center TRUS frames without viewing reference contours. Inter-observer box positioning achieved a mean overlap IoU of 0.736 ± 0.134 .

As detailed in Fig. 8, Observer 2 (more experienced) placed more accurate bounding boxes relative to the expert reference (0.798 ± 0.064 vs. 0.683 ± 0.130 Box IoU), which correspondingly led to higher downstream segmentation performance (0.792 ± 0.081 vs. 0.736 ± 0.142 Dice). In terms of operational efficiency, Observer 2 required less annotation time per frame (4.1 ± 1.2 s vs. 5.6 ± 3.2 s). Notably, even under the less accurate prompts from Observer 1, MIRAUS maintained robust segmentation output (0.736 Dice).

Given the intrinsic anatomical ambiguity and severe boundary noise in prostate TRUS—where inter-expert human agreement itself inherently exhibits noticeable variation [22]—these results highlight the practical resilience of MIRAUS. Even with coarse, operator-dependent bounding boxes, the framework delivers stable segmentation performance while maintaining a rapid, clinically practical annotation workflow.

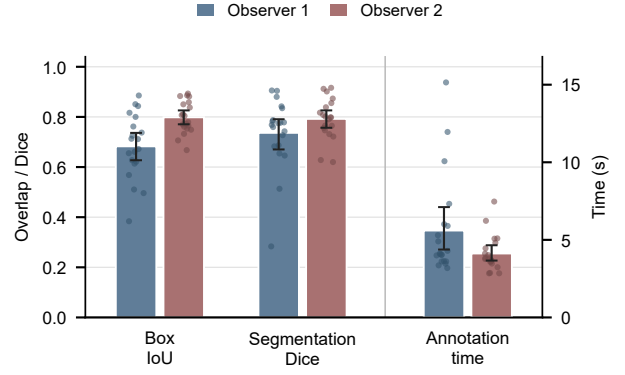


Figure 8: Quantitative comparison of inter-operator variability during box-assisted deployment across two observers with differing clinical experience. Metrics evaluate Box IoU against expert reference, resulting segmentation Dice, and annotation time.

Ellipsoid Volume Measurement Integration To evaluate whether MIRAUS can seamlessly automate traditional caliper-based reporting while addressing operator variability, we benchmarked three volume estimation strategies against the expert manual 3D segmentation reference (V_{ref}) across 73 patients (Fig. 9): (i) **Automated 3D Direct Segmentation** (V_{model}^{3D}), calculated by direct voxel integration over predicted 3D boundaries; (ii) **Model-Assisted Ellipsoid** ($V_{\text{model}}^{\text{ellip}}$), calculated via Eq. (2) using model-derived in-plane diameters paired with the reference longitudinal diameter on a GT-selected axial frame; and (iii) **GT-Mask-Derived Ellipsoid** ($V_{\text{GT}}^{\text{ellip}}$), calculated using the same equation with standardized calipers extracted directly from ground-truth masks manually delineated by an expert.

As shown in Fig. 9, direct 3D automatic segmentation achieved the highest volumetric accuracy against V_{ref} (Bias: +0.60 mL, MAE: 3.61 mL). When evaluating traditional 2D ellipsoid approximations, MIRAUS-derived automated calipers ($V_{\text{model}}^{\text{ellip}}$) achieved a volume estimation bias of +4.80 mL (MAE: 7.50 mL, MAPE: 13.43%). Benchmarking against the GT-derived upper-bound baseline ($V_{\text{GT}}^{\text{ellip}}$: +5.48 mL bias, MAE: 6.28 mL, MAPE: 11.82%) reveals two critical insights: First, the overestimation bias partly reflects deviation of the prostate from the ideal ellipsoidal geometry assumed by the conventional formula, highlighting the fundamental geometric limitation of conventional ellipsoid formulas. Second, replacing GT in-plane dimensions with model-predicted calipers introduced minimal additional error (signed bias: -0.68 mL, MAE: 4.68 mL between $V_{\text{model}}^{\text{ellip}}$ and $V_{\text{GT}}^{\text{ellip}}$). Because ellipsoid volume is multiplicative in the three diameters, errors in the transverse and anteroposterior measurements propagate jointly into the derived volume. With the reference superior-inferior diameter held fixed, the 4.68-mL MAE relative to the reference-caliper ellipsoid

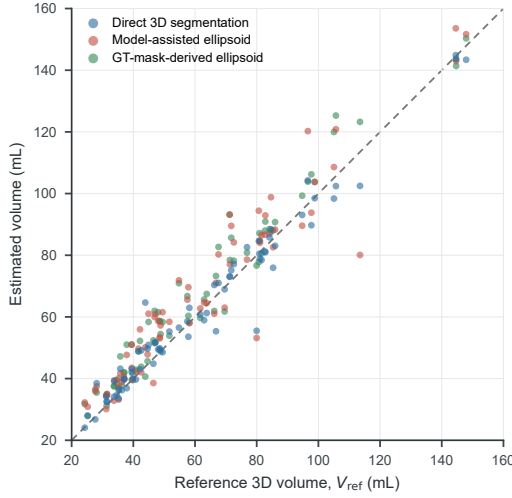


Figure 9: Volumetric estimation accuracy and clinical workflow integration across 73 patients. Comparison among automated 3D direct segmentation ($V_{\text{model}}^{\text{3D}}$), model-assisted ellipsoid estimation ($V_{\text{model}}^{\text{ellip}}$), and GT-mask-derived ellipsoid estimation ($V_{\text{GT}}^{\text{ellip}}$) against the expert manual 3D segmentation gold standard (V_{ref}).

isolates the volumetric effect of replacing the two reference in-plane calipers with model-derived measurements. Because $\text{PSAD} = \text{PSA}/V$, volume overestimation produces a corresponding underestimation of PSAD when the PSA value is fixed.

Overall, MIRAUS unifies boundary estimation with the conventional caliper-based clinical reporting workflow. With sub-15 ms inference enabling rapid processing, our retrospective caliper experiment demonstrates that model-estimated in-plane diameters lessen reliance on manual caliper positioning.

5.3. Limitations and Future Directions

While MIRAUS demonstrates strong efficacy and clinical utility, several methodological scope boundaries highlight promising avenues for future research.

First, regarding anatomical and diagnostic granularity, the current framework primarily focuses on whole-gland prostate segmentation. In advanced targeted biopsy or focal therapy, delineating intra-prostatic malignant lesions, for example PI-RADS ≥ 3 targets, and managing severe tissue deformation from benign prostatic hyperplasia (BPH) remain critical. A crucial next step is to extend the privileged learning formulation to multi-task architectures that simultaneously distill spatial gland boundaries and focal lesion priors from multi-parametric MRI.

Second, the present volume and operator analyses are retrospective, and the external cohort is assessed at the 2D frame level. Although these findings establish technical segmentation and measurement performance across acquisition conditions, their downstream clinical impact, including associations with PSA density, biopsy yield, focal treatment planning, and real-time workflow efficiency, remains to be

examined. Prospective multi-center studies across diverse ultrasound vendors will be essential to translate these technical gains into measurable clinical utility.

Third, regarding multi-slice spatial consistency, processing independent 2D frames offers distinct speed advantages for real-world deployment, yet it does not explicitly enforce inter-slice contour continuity. When extending the model to continuous multi-planar sweeps or 3D volume reconstruction, frame-by-frame inference may introduce inter-slice jitter or boundary fluctuations. Future extensions will explore 3D temporal-spatial smoothing, volumetric tracking, or joint multi-slice attention to guarantee continuous 3D spatial coherence during dynamic scanning.

6. Conclusion

We proposed MIRAUS, a novel multimodal privileged learning framework to improve 2D TRUS prostate boundary segmentation without extra imaging or computational overhead at deployment. Built under the LUPi paradigm, MIRAUS uses multi-slice MRI representations exclusively during training to address speckle noise, low contrast and anatomical ambiguity inherent to TRUS. Its central contribution decouples multimodal supervision from clinical inference: utility-supervised 2.5D aggregation selects informative anatomy from weakly aligned MRI slices, and residual privileged distillation transfers MRI-conditioned feature increments into a lightweight TRUS student. The model does not require MRI or cross-modal registration at test time and supports automatic and box-assisted inference via shared mixed prompts. Rather than relying on challenging dense 3D deformable co-registration, our framework operates with coarse slice-rank mapping to derive distillable anatomical priors from MRI. Mixed-prompt training further endows this dual-mode inference with enhanced robustness against domain shifts.

Evaluations on 73 μRegPro patients show superior automatic segmentation performance (93.20% Dice, 3.62 mm HD95) and low diameter/ellipsoid volume errors in retrospective caliper simulation. On an external cross-center cohort, automatic performance degrades, but box-assisted inference recovers Dice to 84.72%. Our measurement-centric evaluation covers boundary distance, caliper error, volume consistency and inference latency, not merely segmentation overlap.

Collectively, these results validate a key design principle: richer diagnostic modalities can enhance learned anatomical representations without becoming a deployment requirement. This work paves the way for prospective multi-center validation of automatic view selection, full three-diameter volumetry and downstream PSAD assessment.

CRedit authorship contribution statement

Zeming Huang: Conceptualization, Methodology, Software, Formal analysis, Validation, Visualization, Writing - original draft. **Haixin Chen:** Investigation, Resources, Data curation, Validation, Writing - review and editing.

Chuanzhen Cao: Investigation, Resources, Data curation, Writing - review and editing. **Tao Wang:** Conceptualization, Methodology, Supervision, Project administration, Funding acquisition, Writing - review and editing. **Zhipei Huang:** Methodology, Validation, Writing - review and editing. **Fei Qin:** Supervision, Project administration, Writing - review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The public μ RegPro dataset used in this study is available from Zenodo (version 1.1.0; <https://doi.org/10.5281/zenodo.8004388>). The self-collected TRUS cohort contains human-participant imaging data and cannot be made publicly available because the participant consent and institutional ethics approval do not permit unrestricted distribution. Method-verification source code and the five-fold patient assignments are publicly available at <https://github.com/Zeming-Huang/MIRAUS>. Representative predictions and visualizations are available at <https://zeming-huang.github.io/MIRAUS/>.

Funding

This work was supported by the Beijing Municipal Natural Science Foundation under Grant L255012.

References

- [1] F. Bray, M. Laversanne, H. Sung, J. Ferlay, R. L. Siegel, I. Soerjomataram, A. Jemal, Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries, *CA: A Cancer Journal for Clinicians* 74 (3) (2024) 229–263.
- [2] M. S. Litwin, H.-J. Tan, The Diagnosis and Treatment of Prostate Cancer: A Review, *JAMA* 317 (24) (2017) 2532–2542.
- [3] G. Aus, S. Bergdahl, P. Lodding, H. Lilja, J. Hugosson, Prostate Cancer Screening Decreases the Absolute Risk of Being Diagnosed with Advanced Prostate Cancer—Results from a Prospective, Population-Based Randomized Controlled Trial, *European Urology* 51 (3) (2007) 659–664.
- [4] US Preventive Services Task Force, Screening for Prostate Cancer: US Preventive Services Task Force Recommendation Statement, *JAMA* 319 (18) (2018) 1901–1913.
- [5] S. W. D. Merriel, L. Pocock, E. Gilbert, S. Creavin, F. M. Walter, A. Spencer, W. Hamilton, Systematic review and meta-analysis of the diagnostic accuracy of prostate-specific antigen for the detection of prostate cancer in symptomatic patients, *BMC Medicine* 20 (1) (2022) 54.
- [6] S. Loeb, M. A. Bjurlin, J. Nicholson, T. L. Tammela, D. F. Penson, H. B. Carter, P. Carroll, R. Etzioni, Overdiagnosis and Overtreatment of Prostate Cancer, *European Urology* 65 (6) (2014) 1046–1055.
- [7] A. Stanzione, A. Ponsiglione, G. A. Di Fiore, S. G. Picchi, M. Di Stasi, F. Verde, M. Petretta, M. Imbriaco, R. Cuocolo, Prostate volume estimation on MRI: Accuracy and effects of ellipsoid and bullet-shaped measurements on PSA density, *Academic Radiology* 28 (8) (2021) e219–e226.
- [8] R. Colvin, D. Walker, J. Hafron, B. Seifman, S. Nandalur, D. Gangwish, K. R. Nandalur, Which measurement method should be used for prostate volume for PI-RADS? A comparison of ellipsoid and segmentation methods, *Clinical Imaging* 80 (2021) 454–458.
- [9] S. Tong, H. N. Cardinal, R. F. McLoughlin, D. B. Downey, A. Fenster, Intra- and inter-observer variability and reliability of prostate volume measurement via two-dimensional and three-dimensional ultrasound imaging, *Ultrasound in Medicine & Biology* 24 (5) (1998) 673–681.
- [10] L. M. Kurucz, T. Natali, M. Fusaglia, B. Dashtbozorg, Advances in Deep Learning Methods for Prostate Segmentation and Volume Estimation in Ultrasound Imaging, *Applied Sciences* 14 (15) (2024) 6550.
- [11] L. Lenfant, C. Beitone, J. Troccaz, G. Fiard, B. Malavaud, S. Voros, P. C. Mozer, Defining ground truth for prostate segmentation of transrectal ultrasound images: Inter- and intra-observer variability of manual versus semi-automatic methods, *Medical Physics* 52 (8) (2025) e18025.
- [12] Q. Zeng, G. Samei, D. Karimi, C. Kesch, S. S. Mahdavi, P. Abolmaesumi, S. E. Salcudean, Prostate segmentation in transrectal ultrasound using magnetic resonance imaging priors, *Int J Comput Assist Radiol Surg* 13 (6) (2018) 749–757.
- [13] W. L. Smith, C. Lewis, G. Bauman, G. Rodrigues, D. D’Souza, R. Ash, D. Ho, V. Venkatesan, D. Downey, A. Fenster, Prostate volume contouring: A 3D analysis of segmentation using 3DTRUS, CT, and MR, *International Journal of Radiation Oncology*Physics*Physics* 67 (4) (2007) 1238–1247.
- [14] A. D. R. Grey, M. J. Connor, J. Tam, T. Loch, Can transrectal prostate ultrasound compete with multiparametric MRI in the detection of clinically significant prostate cancer?, *Translational Andrology and Urology* 9 (3) (2020) 1492–1500.
- [15] V. Vapnik, R. Izmailov, Learning using privileged information: Similarity control and knowledge transfer, *Journal of Machine Learning Research* 16 (1) (2015) 2023–2049.
- [16] W. Wang, B. Pan, Y. Ai, G. Li, Y. Fu, Y. Liu, LightCM-PNet: A lightweight pyramid network for real-time prostate segmentation in transrectal ultrasound, *Pattern Recognition* 156 (2024) 110776.
- [17] J. Ma, Y. He, F. Li, L. Han, C. You, B. Wang, Segment anything in medical images, *Nature Communications* 15 (1) (2024) 654.
- [18] S. Ghose, A. Oliver, R. Martí, X. Lladó, J. C. Vilanova, J. Freixenet, J. Mitra, D. Sidibé, F. Meriaudeau, A survey of prostate segmentation methodologies in ultrasound, magnetic resonance and computed tomography images, *Comput. Methods Programs Biomed.* 108 (1) (2012) 262–287.
- [19] L. Gong, S. Pathak, D. Haynor, P. Cho, Y. Kim, Parametric shape modeling using deformable superellipses for prostate segmentation, *IEEE Transactions on Medical Imaging* 23 (3) (2004) 340–349.
- [20] X. Li, C. Li, A. Fedorov, T. Kapur, X. Yang, Segmentation of prostate from ultrasound images using level sets on active band and intensity variation across edges, *Med Phys* 43 (6) (2016) 3090–3103.
- [21] E. M. A. Anas, P. Mousavi, P. Abolmaesumi, A deep learning approach for real time prostate segmentation in freehand ultrasound guided biopsy, *Med Image Anal* 48 (2018) 107–116.
- [22] N. Ghavami, Y. Hu, E. Bonmati, R. Rodell, E. D. Gibson, C. M. Moore, D. C. Barratt, Automatic slice segmentation of intraoperative transrectal ultrasound images using convolutional neural networks, in: R. J. Webster, B. Fei (Eds.), *Medical Imaging 2018: Image-Guided Procedures, Robotic Interventions, and Modeling*, SPIE, Houston, United States, 2018, p. 2.
- [23] Ö. Çiçek, A. Abdulkadir, S. S. Lienkamp, T. Brox, O. Ronneberger, 3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation, in: S. Ourselin, L. Joskowicz, M. R. Sabuncu, G. Unal, W. Wells (Eds.), *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2016*, Springer International Publishing, Cham, 2016, pp. 424–432.
- [24] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, K. H. Maier-Hein, nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation, *Nature Methods* 18 (2) (2021) 203–211.

- [25] N. Ghavami, Y. Hu, E. Bonmati, R. Rodell, E. Gibson, C. Moore, D. Barratt, Integration of spatial information in convolutional neural networks for automatic segmentation of intraoperative transrectal ultrasound images, *J. Med. Imaging* 6 (1) (2019) 011003.
- [26] Y. Wang, H. Dou, X. Hu, L. Zhu, X. Yang, M. Xu, J. Qin, P.-A. Heng, T. Wang, D. Ni, Deep Attentive Features for Prostate Segmentation in 3D Transrectal Ultrasound, *IEEE Trans Med Imaging* 38 (12) (2019) 2768–2778.
- [27] N. Orlando, D. J. Gillies, I. Gyacskov, C. Romagnoli, D. D’Souza, A. Fenster, Automatic prostate segmentation using deep learning on clinically diverse 3D transrectal ultrasound images, *Med Phys* 47 (6) (2020) 2413–2426.
- [28] R. J. Van Sloun, R. R. Wildeboer, C. K. Mannaerts, A. W. Postema, M. Gayet, H. P. Beerlage, G. Salomon, H. Wijkstra, M. Misch, Deep Learning for Real-time, Automatic, and Scanner-adapted Prostate (Zone) Segmentation of Transrectal Ultrasound, for Example, Magnetic Resonance Imaging–transrectal Ultrasound Fusion Prostate Biopsy, *European Urology Focus* 7 (1) (2021) 78–85.
- [29] S. Vesal, I. Gayo, I. Bhattacharya, S. Natarajan, L. S. Marks, D. C. Barratt, R. E. Fan, Y. Hu, G. A. Sonn, M. Rusu, Domain Generalization for Prostate Segmentation in Transrectal Ultrasound Images: A Multi-center Study, *Med. Image Anal.* 82 (2022) 102620.
- [30] X. Xu, T. Sanford, B. Turkbey, S. Xu, B. J. Wood, P. Yan, Shadow-consistent Semi-supervised Learning for Prostate Ultrasound Segmentation, *IEEE transactions on medical imaging* 41 (6) (2022) 1331–1345.
- [31] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, P. Dollár, R. Girshick, Segment Anything, in: 2023 IEEE/CVF International Conference on Computer Vision (ICCV), 2023, pp. 3992–4003.
- [32] K. Zhang, D. Liu, Customized segment anything model for medical image segmentation (2023). [arXiv:2304.13785](https://arxiv.org/abs/2304.13785).
- [33] J. Cheng, J. Ye, Z. Deng, J. Chen, T. Li, H. Wang, Y. Su, Z. Huang, J. Chen, L. Jiang, H. Sun, J. He, S. Zhang, M. Zhu, Y. Qiao, Sammed2d (2023). [arXiv:2308.16184](https://arxiv.org/abs/2308.16184).
- [34] X. Lin, Y. Xiang, L. Yu, Z. Yan, Beyond adapting sam: Towards end-to-end ultrasound image segmentation via auto prompting (2024). [arXiv:2309.06824](https://arxiv.org/abs/2309.06824).
- [35] K. Wu, J. Zhang, H. Peng, M. Liu, B. Xiao, J. Fu, L. Yuan, TinyViT: Fast Pretraining Distillation for Small Vision Transformers, *arXiv:2207.10666* (Jul. 2022). [arXiv:2207.10666](https://arxiv.org/abs/2207.10666).
- [36] Y. Du, F. Bai, T. Huang, B. Zhao, Segvol: Universal and interactive volumetric medical image segmentation, in: A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, C. Zhang (Eds.), *Advances in Neural Information Processing Systems*, Vol. 37, Curran Associates, Inc., 2024, pp. 110746–110783.
- [37] F. Isensee, M. Rokuss, L. Krämer, S. Dinkelacker, A. Ravindran, F. Stritzke, B. Hamm, T. Wald, M. Langenberg, C. Ulrich, J. Deissler, R. Floca, K. Maier-Hein, nninteractive: Redefining 3d promptable segmentation (2025). [arXiv:2503.08373](https://arxiv.org/abs/2503.08373).
- [38] D. Lopez-Paz, L. Bottou, B. Schölkopf, V. Vapnik, Unifying distillation and privileged information (2016). [arXiv:1511.03643](https://arxiv.org/abs/1511.03643).
- [39] J. Hoffman, S. Gupta, T. Darrell, Learning With Side Information Through Modality Hallucination, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 826–834.
- [40] J. Lambert, O. Sener, S. Savarese, Deep Learning Under Privileged Information Using Heteroscedastic Dropout, in: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, IEEE, Salt Lake City, UT, 2018, pp. 8886–8895.
- [41] C. Chen, Q. Dou, Y. Jin, Q. Liu, P. A. Heng, Learning With Privileged Multimodal Knowledge for Unimodal Segmentation, *IEEE Transactions on Medical Imaging* 41 (3) (2022) 621–632.
- [42] Z. Gao, J. Chung, M. Abdelrazek, S. Leung, W. K. Hau, Z. Xian, H. Zhang, S. Li, Privileged Modality Distillation for Vessel Border Detection in Intracoronary Imaging, *IEEE Trans. Med. Imaging* 39 (5) (2020) 1524–1534.
- [43] M. Hu, M. Maillard, Y. Zhang, T. Ciceri, G. La Barbera, I. Bloch, P. Gori, Knowledge Distillation from Multi-modal to Mono-modal Segmentation Networks, in: A. L. Martel, P. Abolmaesumi, D. Stoyanov, D. Mateus, M. A. Zuluaga, S. K. Zhou, D. Racocanu, L. Joskowicz (Eds.), *Medical Image Computing and Computer Assisted Intervention – MICCAI 2020*, Springer International Publishing, Cham, 2020, pp. 772–781.
- [44] Y. Ding, X. Yu, Y. Yang, RFNet: Region-aware fusion network for incomplete multi-modal brain tumor segmentation, in: *Proc. IEEE/CVF International Conference on Computer Vision*, 2021, pp. 3975–3984.
- [45] Y. Zhang, N. He, J. Yang, Y. Li, D. Wei, Y. Huang, Y. Zhang, Z. He, Y. Zheng, mmFormer: Multimodal medical transformer for incomplete multimodal learning of brain tumor segmentation, in: *Medical Image Computing and Computer Assisted Intervention*, 2022, pp. 107–117.
- [46] S. Wang, Z. Yan, D. Zhang, H. Wei, Z. Li, R. Li, Prototype knowledge distillation for medical segmentation with missing modality, in: *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing*, 2023, pp. 1–5.
- [47] Y. Fu, Y. Lei, T. Wang, P. Patel, A. B. Jani, H. Mao, W. J. Curran, T. Liu, X. Yang, Biomechanically constrained non-rigid MR-TRUS prostate registration using deep learning based 3D point cloud matching, *Med Image Anal* 67 (2021) 101845.
- [48] H. Guo, M. Kruger, S. Xu, B. J. Wood, P. Yan, Deep adaptive registration of multi-modal prostate images, *Comput Med Imaging Graph* 84 (2020) 101769.
- [49] Q. Dou, Q. Liu, P. A. Heng, B. Glocker, Unpaired Multi-Modal Segmentation via Knowledge Distillation, *IEEE Trans. Med. Imaging* 39 (7) (2020) 2415–2425.
- [50] Z. M. C. Baum, S. U. Saeed, Z. Min, Y. Hu, D. C. Barratt, MR to ultrasound registration for prostate challenge - dataset (Jun. 2023).
- [51] A. Shaker, M. Maaz, H. Rasheed, S. Khan, M.-H. Yang, F. Shahbaz Khan, UNETR++: Delving Into Efficient and Accurate 3D Medical Image Segmentation, *IEEE Transactions on Medical Imaging* 43 (9) (2024) 3377–3390.
- [52] S. Kornblith, M. Norouzi, H. Lee, G. Hinton, Similarity of neural network representations revisited, in: K. Chaudhuri, R. Salakhutdinov (Eds.), *Proceedings of the 36th International Conference on Machine Learning*, Vol. 97 of *Proceedings of Machine Learning Research*, PMLR, 2019, pp. 3519–3529.